



De Mathematics Competitions

1st Annual

DMC 8

Sunday, November 3, 2024



INSTRUCTIONS

1. DO NOT OPEN THIS BOOKLET UNTIL YOUR COMPETITION MANAGER TELLS YOU TO BEGIN.
2. This is a 25-question multiple choice test. For each question, only one answer choice is correct.
3. Mark your answer to each problem on the answer sheet with a #2 pencil. Check blackened answers for accuracy and erase errors completely. Only answers that are properly marked on the answer sheet will be scored.
4. SCORING: You will receive 1 point for each correct answer, 0 points for each problem left unanswered, and 0 points for each incorrect answer.
5. Only blank scratch paper, rulers, and erasers are allowed as aids. Prohibited materials include calculators, smart-watches, phones, computing devices, compasses, protractors, and graph paper. No problems on the competition will require the use of a calculator.
6. Figures are not necessarily drawn to scale.
7. You will have 40 minutes to complete the competition once you start the test.

The problems and solutions for this DMC 8 were prepared
by the DMC Committee under the direction of:

Taiki Aiba and Ryan Shin

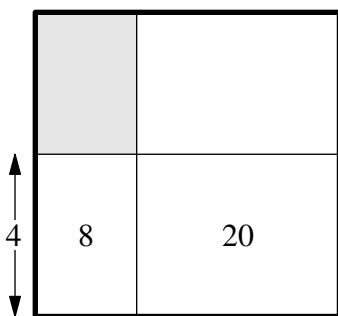
The DMC Committee reserves the right to disqualify scores if it determines that the rules or the required security procedures were not followed.

The publication, reproduction or communication of the problems or solutions of this competition during the period when students are eligible to participate seriously jeopardizes the integrity of the results. Dissemination via phone, email, or digital media of any type during this period is a violation of the competition rules.

1. What is the tens digit of

$$20,252,025 + 20,002,500 + 2,025 + 20 + 25?$$

- (A) 5 (B) 6 (C) 7 (D) 8 (E) 9
2. Which of the following numbers would increase when rounded to the nearest integer?
- (A) -52.8 (B) -7.3 (C) 0.1 (D) 41.4 (E) 83.2
3. At the age of 7, Nayeon borrowed a book from the library, but she lost the book. Every year that the book is missing, Nayeon is charged \$1.50. By the time she finds the book and returns it to the library, she has paid \$28.50. How old is Nayeon at the time of returning the book?
- (A) 19 (B) 20 (C) 24 (D) 25 (E) 26
4. Ladha splits a square into 4 smaller rectangles by drawing two lines parallel to the sides of the rectangle, as shown below. Two of the smaller rectangles are labeled with their areas, and the height of the bottom two rectangles is 4. What is the area of the shaded rectangle?



- (A) 4 (B) 6 (C) 8 (D) 9 (E) 12
5. The frequency and wavelength of an electromagnetic wave are related by the equation below, where frequency is measured in Hertz, and wavelength is measured in meters.

$$2.998 \times 10^8 = (\text{frequency}) \times (\text{wavelength})$$

The wavelength of an electromagnetic wave is 5.0×10^{-2} meters. Which of the following is closest to the frequency of the wave, in Hertz?

- (A) 1.7×10^{-10} (B) 1.7×10^9 (C) 6.0×10^9 (D) 1.7×10^{10} (E) 6.0×10^{10}

6. Of the students at Ken's school, two-thirds are seventh graders and one-third are eighth graders. Two-fifths of seventh graders and one-fourth of eighth graders like math. What fraction of the students that like math are eighth graders?

(A) $\frac{1}{12}$ (B) $\frac{1}{5}$ (C) $\frac{5}{21}$ (D) $\frac{4}{15}$ (E) $\frac{2}{3}$

7. The *deviation* of a number in a list is defined as

$$(\text{deviation}) = (\text{number}) - (\text{mean}).$$

Audrey finds the sum of the deviations of all the numbers in the list 2, 3, 5, 8, 10, 12, 19, 23, and 35. What is this sum?

(A) -12 (B) 0 (C) 4 (D) 10 (E) 13

8. Jessica, Julia, and Jennifer are playing a game. Each of them starts with 0 points. Each round, the winner gets to steal 3 points from another person and give it to themselves, and the person in second place gets 1 point. At the end of the game, the sum of the number of points of Jessica, Julia, and Jennifer is 12. How many rounds were played?

(A) 2 (B) 3 (C) 4 (D) 6 (E) 12

9. Let the letters D , M , and C represent digits. Billy writes the true equation

$$\underline{D} \, \underline{2} \, \underline{M} \times 3 = \underline{M} \, \underline{7} \, \underline{C}.$$

What is the value of $D + M + C$?

(A) 7 (B) 9 (C) 12 (D) 13 (E) 16

10. When Patrick wrote the sum of the numbers from 2 to 9, inclusive, he accidentally added one of the numbers n times in total, resulting in the number 99. What is the value of n ?

(A) 2 (B) 5 (C) 7 (D) 12 (E) 13

11. The video game Super Mario Bros. runs at 60.08 frames per second. Each frame of Super Mario Bros. runs for m minutes. Which of the following is true about the value of m ?

(A) $0 < m < \frac{1}{3600}$ (B) $\frac{1}{3600} < m < \frac{1}{1200}$ (C) $\frac{1}{1200} < m < \frac{1}{600}$
(D) $\frac{1}{600} < m < \frac{1}{60}$ (E) $\frac{1}{60} < m < 1$

12. Ryan rolls a standard 6-sided die. What is the fewest possible number of rolls he could make so that the product of all the numbers rolled ends with 13 zeroes?

(A) 13 (B) 14 (C) 19 (D) 20 (E) 26

13. Two cups of different sizes are filled with a mix of water and juice. The liquid in one cup is 20% juice, and the liquid in the other cup is 50% juice. Upon pouring the contents of both cups into an empty pitcher, the liquid in the pitcher is 44% juice. What is the ratio of the size of the larger cup to the size of the smaller cup?

(A) 2 : 1 (B) 3 : 1 (C) 4 : 1 (D) 5 : 1 (E) 6 : 1

14. David is playing the game 24. He wants to pick integers a and b so that the equations

$$(a + 2) \div (b - 4) = 24$$

$$(a + 2) \times (b - 4) = 24$$

are satisfied. In how many ways can he pick the values of a and b ?

(A) 0 (B) 1 (C) 2 (D) 3 (E) 4

15. Ian's favorite number has p positive factors. When he multiplies it by 49, the resulting number has $\frac{4}{3}p$ positive factors. Suppose that he divides it by 7 instead. How many positive factors will the resulting number have?

(A) $\frac{1}{2}p$ (B) $\frac{2}{3}p$ (C) $\frac{3}{4}p$ (D) $\frac{5}{6}p$ (E) the result is not an integer

16. Kazuha has 8 distinct numbers, 5 of which are negative and 3 of which are positive. How many ways can Kazuha choose a group with at least one of these numbers so that the product of the numbers in the group is positive?

(A) 63 (B) 64 (C) 119 (D) 127 (E) 255

17. On a test, the median score of 10 students is 85. The highest student's score is removed. Which of the following statements are necessarily true?

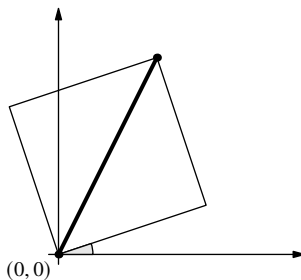
I. If the new median is less than 85, then one of the test scores is 85.

II. If the new median is not 85, then none of the test scores are 85.

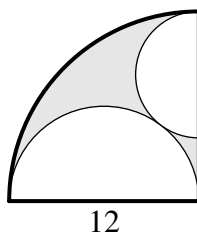
III. If one of the test scores is 85, then the new median is 85.

(A) I, II, and III (B) I only (C) I and II only (D) III only (E) II and III only

18. A square has a vertex at the origin. The diagonal of the square containing the origin has slope $\sqrt{3}$. What is the measure of the angle between the square's side and the positive x -axis?



- (A) 12° (B) 15° (C) 18° (D) 20° (E) 30°
19. How many ways can the first nine positive integers be colored red, green, and yellow such that no two integers that leave the same remainder when divided by 3 are the same color?
- (A) 180 (B) 216 (C) 360 (D) 432 (E) 720
20. A semicircle is drawn inside a quarter circle with radius 12 such that its diameter is shared with a radius of the quarter circle, and another semicircle is drawn so that its diameter is on the other radius of the quarter circle, and it touches the other semicircle at exactly one point.



What is the area of the region inside the quarter circle and outside both semicircles?

- (A) 9π (B) 10π (C) 11π (D) 12π (E) 13π
21. What is the sum of the digits of the largest multiple of 3 with the property that the digits of a smaller multiple of 3 do *not* appear in a row in that number? (For example, 1203 does not have this property because the digits of 0, 3, 12, and 120 appear in a row.)
- (A) 24 (B) 27 (C) 30 (D) 33 (E) 36

22. Students Arn, Bob, Cyd, Dan, and Eve are arranged in that order in a circle. They start counting: Arn first, then Bob, and so forth. Arn says the number 121, after which each person says a number either 4 or 6 more than the previous number. At the end, Bob says the number 169. How many times did a student say a number 4 more than the previous number?

(A) 2 (B) 6 (C) 9 (D) 11 (E) 16

23. David has six blocks arranged in the following order.

A B C D E F

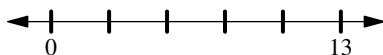
Every move, he chooses two of the blocks and switches their positions. After three moves, he ends up with the following arrangement.

C D E B A F

How many possible sequences of three moves are there?

(A) 3 (B) 6 (C) 8 (D) 9 (E) 12

24. On a number line, Sophia draws 6 equally-spaced ticks from 0 to 13, as shown below.



Jason draws n equally-spaced ticks from 0 to 13. The shortest *positive* distance between one of Sophia's ticks and one of Jason's ticks is $\frac{1}{5}$. What is the sum of all possible values of n ?

(A) 17 (B) 68 (C) 69 (D) 80 (E) 82

25. Pedro writes either 1 or 3 in every cell of a 6×6 grid so that the product of the numbers in each column has a units digit of 9, and the product of the numbers in each row has a units digit of 1. He writes as many threes as possible. In how many ways can he fill the grid?

(A) 360 (B) 720 (C) 840 (D) 1200 (E) 1800